OpenClaw AI Voice Interaction Configuration

1. Course Overview

This course introduces the international version configuration method for OpenClaw AI voice interaction in the Jetson Nano Docker container, focusing on English ASR, English TTS, international regional settings, local Ollama large language model, and OpenRouter online LLM configuration.

After completing this course, you will be able to:

- Know which configuration files need to be modified for international voice interaction
- Switch speech recognition, model interface, and speech synthesis to English
- Switch between Ollama and OpenRouter large language model platforms
- Start and verify the English voice interaction pipeline

2. Voice Interaction Pipeline

The international voice interaction flow is as follows:

```
Microphone
  -> ASR speech recognition node
  -> /asr
  -> OpenClaw Bridge
  -> openclaw agent
  -> Large model reply
  -> /tts_text_input
  -> TTS node
  -> Speaker playback
```

Main code directory:

```
/root/yahboom_ws/src/largemodel
```

Main configuration files:

```
/root/yahboom_ws/src/largemodel/config/yahboom.yaml
/root/yahboom_ws/src/largemodel/config/large_model_interface.yaml
```

3. Enter Docker Container

To start a new container:

```
~/run_usb_docker.sh
```

Or if a container is already running, enter it with:

```
~/enter_docker.sh
```

Check if any containers are running:

```
docker ps
```

4. Configure `yahboom.yaml`

Configuration file:

```
/root/yahboom_ws/src/largemodel/config/yahboom.yaml
```

For international version, you need to enable the `International Edition` configuration block and comment out or remove the China edition configuration block with the same names to avoid duplication of `asr`, `model_service`, and `tts_only_node`.

Recommended configuration:

```
###############---------------------------------------------International Edition-----------
-----------------------------------------#################
asr:
```

```yaml
  ros__parameters:
    VAD_MODE: 2
    sample_rate: 16000
    frame_duration_ms: 30
    use_oline_asr: False
    mic_serial_port: "/dev/ttyUSB0"
    mic_index: 11
    language: 'en'
    regional_setting: "international"

model_service:
  ros__parameters:
    language: 'en'
    useolinetts: False
    llm_platform: 'ollama'
    # Available platforms: 'ollama', 'openrouter'
    regional_setting: "international"
    camera_width: 640
    camera_height: 480
    camera_type: 'csi'

tts_only_node:
  ros__parameters:
    language: 'en'
    useolinetts: False
    regional_setting: "international"
```

Key parameter descriptions:

- `language: 'en'`: Use English for ASR, LLM interface, and TTS
- `regional_setting: "international"`: Enable international edition logic
- `use_oline_asr: False`: Use offline English ASR
- `useolinetts: False`: Use offline English TTS
- `llm_platform: 'ollama'`: Use local Ollama LLM
- `llm_platform: 'openrouter'`: Use OpenRouter online LLM

> Note: If the upper part of the current file still has China edition configuration, you must first comment out the China edition before enabling the international edition. Otherwise, the actual runtime may still read China edition parameters.

5. Configure `large_model_interface.yaml`

Configuration file:

```
/root/yahboom_ws/src/largemodel/config/large_model_interface.yaml
```

5.1 Local Ollama LLM

If using in `yahboom.yaml`:

```
llm_platform: 'ollama'
```

Then check the Ollama configuration:

```
ollama_host: "http://localhost:11434"
ollama_model: "llava-phi3:latest"
```

Verify Ollama service:

```
curl http://localhost:11434/api/tags
```

5.2 OpenRouter Online LLM

If you need an international online LLM, change `yahboom.yaml` to:

```
llm_platform: 'openrouter'
```

Then configure:

```
## -----------------English online large model (English online large model configuration)---
--------
openrouter_api_key: "skxxxxxxxxxxxxxx"
openrouter_model: "qwen/qwen2.5-vl-72b-instruct:free"
```

Parameter descriptions:

- `openrouter_api_key`: Fill in your own OpenRouter API Key
- `openrouter_model`: OpenRouter model name

Network verification:

```
curl -I https://openrouter.ai
```

6. English ASR and TTS

6.1 English Offline ASR

Offline ASR model in `large_model_interface.yaml`:

```
local_asr_model: "/root/yahboom_ws/src/largemodel/MODELS/asr/SenseVoiceSmall"
```

Required settings in `yahboom.yaml`:

```yaml
use_oline_asr: False
language: 'en'
regional_setting: "international"
```

6.2 English Offline TTS

English TTS model in `large_model_interface.yaml`:

```yaml
en_tts_model: "/root/yahboom_ws/src/largemodel/MODELS/tts/en/en_US-libritts-high.onnx"
en_tts_json: "/root/yahboom_ws/src/largemodel/MODELS/tts/en/en_US-libritts-high.onnx.json"
```

Check model files:

```bash
ls -lh /root/yahboom_ws/src/largemodel/MODELS/tts/en/
```

Required settings in `yahboom.yaml`:

```yaml
useolinetts: False
language: 'en'
regional_setting: "international"
```

7. Microphone Configuration

Default configuration:

```yaml
mic_serial_port: "/dev/ttyUSB0"
mic_index: 11
sample_rate: 16000
```

If you see during startup:

```
Microphone check failed for index 11
```

It means the microphone index in the current container may not be `11`. After entering the container, run:

```bash
python3 - <<'PY'
import pyaudio
p = pyaudio.PyAudio()
for i in range(p.get_device_count()):
    d = p.get_device_info_by_index(i)
    print(i, d.get("name"), "in", d.get("maxInputChannels"), "out",
d.get("maxOutputChannels"), "rate", d.get("defaultSampleRate"))
p.terminate()
PY
```

Find the USB microphone index with input channels, then modify:

```yaml
mic_index: <actual_index>
```

If you see:

```
OSError: [Errno -9997] Invalid sample rate
```

It means the microphone does not support the current sample rate. You need to adjust the sample rate according to the device's supported rates.

8. Rebuild After Modification

After modifying the source directory configuration, it is recommended to rebuild:

```
cd /root/yahboom_ws
colcon build --packages-select largemodel
source /root/yahboom_ws/install/setup.bash
```

9. Start Voice Interaction

9.1 Start OpenClaw Gateway

```
openclaw gateway
```

Health check:

```
curl http://127.0.0.1:18789/health
```

Normal output:

```
{"ok":true,"status":"live"}
```

9.2 One-Click Start

```
cd /root/yahboom_ws/src/largemodel
python3 main.py
```

When you see:

```
Voice interaction is running
```

The voice interaction pipeline has been started.

Then use the voice module, wake it with the wake word "Hi,yahboom", and say a test phrase, for example:

- How's the weather today?
- Tell me a joke.
- What time is it now?

11. FAQ

11.1 Still Reading China Edition Configuration

Check if the China edition configuration block is still enabled in `yahboom.yaml`:

```
grep -n "China\|international\|language" /root/yahboom_ws/src/largemodel/config/yahboom.yaml
```

If the China edition in the upper part is not commented out, the program may still run according to the China edition.

11.2 No Response from OpenRouter

Check:

- Whether `llm_platform` is set to `openrouter`
- Whether `openrouter_api_key` is a real key
- Whether `openrouter_model` is available
- Whether Jetson can access OpenRouter

11.3 No Audio Playback

Check:

- Whether `tts_only_node.language` is set to `en`
- Whether `tts_only_node.useolinetts` is set to `False`
- Whether the English TTS model file exists
- Whether the USB sound card or speaker is properly connected

11.4 Microphone Check Failed

Check:

- Whether `ls /dev/ttyUSB*` shows any devices
- Whether `mic_index` is correct
- Whether the microphone is mapped into the Docker container
- Whether the current sample rate is supported by the microphone